

Office Automation Software Lab

Complete student lesson for course code **2259**.

Revision 2026 Semester II Laboratory 4 modules

Course snapshot

Scheme: 0:0:3:0 (L:T:P:R)

Credits: 1.5

Contact: 45 hours

Module hours listed: 51

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

2259

Semester

II

Credits

1.5

Teaching scheme

0:0:3:0 (L:T:P:R)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	File Creation, Organisation, Security and Recovery	9	CO1
2	MS Office Documents, Presentations, Malayalam Typing and Excel	18	CO2
3	Cloud Collaboration, Google Workspace and WPS Compatibility	12	CO3
4	AI Content, Editing, Spreadsheet Automation and Open-ended Work	12	CO4

Prerequisites

- Nil.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. Develop file creation, organisation, and secure handling skills for business data operations.
2. Use MS Word, PowerPoint, introductory Excel, and Malayalam typing tools including ISM and Google Typing.
3. Use Google Docs, Sheets, Slides, MS 365, WPS, and Google Workspace for collaboration.
4. Use AI-based automation tools to enhance office productivity.

Course Outcomes

1. Implement file creation, organisation, and management strategies for secure business storage and retrieval.
2. Create formatted documents, presentations, and basic spreadsheets using MS Office tools with multilingual typing integration.
3. Use cloud tools for real-time collaboration and online document management.
4. Apply AI tools to automate office tasks and promote sustainable, innovative practice.

CO-PO mapping as supplied

CO-PO matrix is reproduced from the supplied syllabus; correlation levels use the source legend where provided.

Legend: 3 = high/strong correlation; 2 = medium/moderate correlation; 1 = low/weak correlation; - = no correlation.

The supplied PDF includes a full CO-PO matrix for CO1-CO4 and PO1-PO11.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	File creation and exploration	1
module-1	M1.02	Basic file organisation	1
module-1	M1.03	File compression and archiving	1
module-1	M1.04	File permissions and access control	2
module-1	M1.05	Backup and recovery strategies	2
module-1	M1.06	File search and filtering techniques	2
module-2	M2.01	MS Word document creation and formatting	3
module-2	M2.02	Mail merge for personalised correspondence	3
module-2	M2.03	Multilingual document creation using ISM	3
module-2	M2.04	PowerPoint slide design and multimedia integration	3
module-2	M2.05	Introductory Excel data entry, formulae and charts	3
module-2	M2.06	Series Exam I and practical record	3
module-3	M3.01	Collaborative document editing in Google Docs	1.5
module-3	M3.02	Collaborative data management in Google Sheets	1.5
module-3	M3.03	Presentation collaboration in Google Slides	1.5
module-3	M3.04	Co-authored files, change tracking and mobile usability	1.5
module-3	M3.05	WPS Office compatibility	1.5
module-3	M3.06	End-to-end collaboration in Google Workspace	3
module-4	M4.01	Content generation using Gemini or ChatGPT	3
module-4	M4.02	AI-assisted editing and summarisation with Grammarly or Copilot	3

Module	Topic code	Topic	Hours
module-4	M4.03	AI automation in spreadsheets and presentations	3
module-4	M4.04	Open-ended experiment and business sustainability	3
module-4	M4.05	Series Exam II and lab examination preparation	3

Learning Roadmap

1. File Creation, Organisation, Security and Recovery
CO1 · 9 hours

2. MS Office Documents, Presentations, Malayalam Typing and Excel
CO2 · 18 hours

3. Cloud Collaboration, Google Workspace and WPS Compatibility
CO3 · 12 hours

4. AI Content, Editing, Spreadsheet Automation and Open-ended Work
CO4 · 12 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

File Creation, Organisation, Security and Recovery

This module develops safe file-handling habits for business data storage and retrieval.

Hours: 9

CO1 · Bloom level: Applying

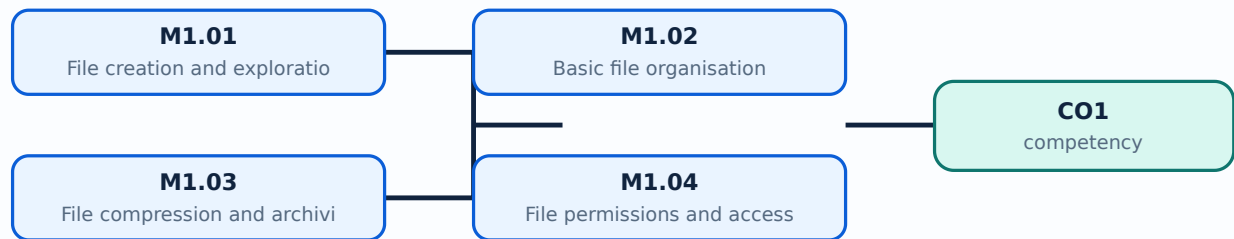
Learning goals

- Create and name files using conventions.
- Organise folders and navigate documented paths.

- Compress, protect, back up, recover, search, and filter files.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

– M1.01 — File creation and exploration (1 hours)

Concept and explanation

Files store documents, data, media, and program outputs. Students should identify file type, extension, location, owner, version, and purpose. Naming conventions use meaningful names, dates, and version labels without unsafe characters or ambiguity.

Malayalam support: File-രൂപം type, name, location, owner, version, purpose ഉൾപ്പെടെ വിവരങ്ങൾ നൽകണം.

Study points

- Create 5-10 different file types.
- Use a consistent naming convention and verify openability.

Suggested procedure

1. Create sample files and record name, type, size, location, and purpose.

Engineering / workplace application: Business records depend on predictable file names and locations.

Illustrative example: `SalesReport\_2026-08\_v01.xlsx` is clearer than `newfile.xlsx`.

Common mistake: Names such as final_final2 give no reliable version information.

– M1.02 — Basic file organisation (1 hours)

Concept and explanation

Folder structures should reflect business function, project, date, and access need. A documented navigation path helps another user find a record without guesswork. Avoid excessive nesting and duplicate copies.

Malayalam support: Folder structure function/project/date ഫയർഫോൾഡർ; navigation path document ഡോക്യുമെന്റ്.

Study points

- Organise a small business database folder.
- Document the path and ownership.

Suggested procedure

1. Create a folder tree and test it with a second user.

Engineering / workplace application: A clear folder structure reduces search time and duplicate work.

Illustrative example: A project folder can contain 01_Source, 02_Working, 03_Approved, and 04_Archive.

Common mistake: Saving every file on the desktop or mixing personal and business data is unsafe.

– M1.03 — File compression and archiving (1 hours)

Concept and explanation

Compression reduces storage or transfer size and archiving preserves a project snapshot. Integrity checks confirm that the archive opens and files are complete. The original should be retained according to policy.

Malayalam support: Compression size കുറയ്ക്കുന്നു; archive integrity check ഉറപ്പുവരുത്തുന്നു; original retention policy പാലിയം.

Study points

- Compress a sample project folder.
- Compare size and verify extraction.

Suggested procedure

1. Compress, record original/archive sizes, extract to a new folder, and compare files.

Engineering / workplace application: Archives support transfer, backup, and long-term record management.

Illustrative example: A project folder can be zipped and restored to a test location.

Common mistake: Deleting the original before verifying the archive risks data loss.

– M1.04 — File permissions and access control (2 hours)

Concept and explanation

Access control determines who may read, modify, share, or delete a file. Least privilege, strong authentication, secure sharing, and access logs protect privacy. Permissions should match role and be reviewed.

Malayalam support: ഏറ്റവും കുറഞ്ഞ അനുമതി; read/modify/share/delete permissions role അനുസരിച്ച് പരിശോധിക്കുക.

Study points

- Create a secured file set with access logs.
- Explain privacy issues and review permissions.

Suggested procedure

1. Assign hypothetical roles and record permitted operations.

Engineering / workplace application: Payroll or customer files require stricter access than a public event poster.

Illustrative example: A trainee may view a template but not delete the approved financial report.

Common mistake: Sharing an edit link publicly exposes data beyond the intended users.

– M1.05 — Backup and recovery strategies (2 hours)

Concept and explanation

Backup copies support recovery from deletion, corruption, hardware failure, or ransomware. A plan states what is backed up, frequency, location, retention, access, and test-recovery procedure. Recovery should be tested rather than assumed.

Malayalam support: Backup plan-എന്തെങ്കിലും, frequency, location, retention, access, restore test പരീക്ഷിക്കുക പരീക്ഷിക്കുക.

Study points

- Prepare a backup plan document.
- Demonstrate recovery of a deleted sample file.

Suggested procedure

1. Delete a dummy file, restore it, and record evidence and limitations.

Engineering / workplace application: Businesses use local and cloud backups with controlled access and periodic restore tests.

Illustrative example: A weekly archive plus daily versioned backup can support recovery depending on policy.

Common mistake: A synchronised copy is not always an independent backup; deletion may synchronise too.

— M1.06 — File search and filtering techniques (2 hours)

Concept and explanation

Search uses file name, type, date, size, content, tags, or location to find records. Filters reduce a large set to relevant files. Search results should be checked for version, authority, and access.

Malayalam support: Search name, type, date, size, content, location ഏകദേശം
ഏകദേശം; result version/authority verify ഏകദേശം.

Study points

- Locate a file in a cluttered folder set.
- Use at least two filters and record the result.

Suggested procedure

1. Create a cluttered sample folder and document a repeatable search method.

Engineering / workplace application: Efficient search reduces time spent on document retrieval and supports audit work.

Illustrative example: Filter `.xlsx` files modified this month in a project archive.

Common mistake: Opening the first similarly named file without checking version can use obsolete data.

Module summary: Good office automation begins with disciplined file names, structure, access, backup, and retrieval.

OFFICIAL MODULE 2

MS Office Documents, Presentations, Malayalam Typing and Excel

This module creates professional business documents and presentations and introduces multilingual typing and spreadsheet calculations.

Hours: 18

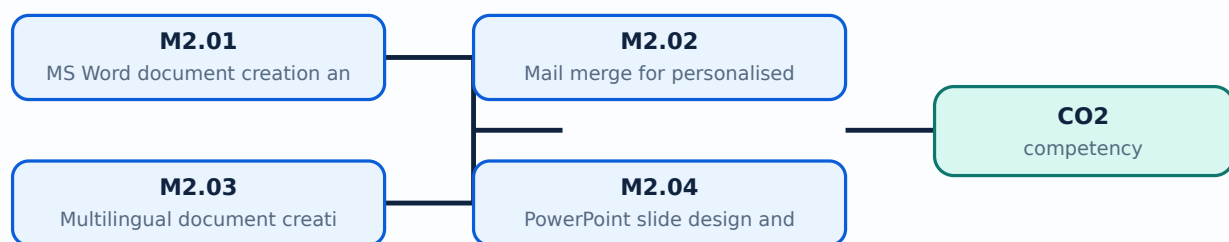
CO2 · Bloom level: Applying

Learning goals

- Create a formatted business report and mail merge.
- Create a multilingual document using ISM.
- Design a PowerPoint presentation and export it.
- Enter introductory Excel data, formulae, and charts.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

– M2.01 — MS Word document creation and formatting (3 hours)

Concept and explanation

A professional Word report uses styles, headings, page layout, tables, headers/footers, page numbers, consistent spacing, proofreading, and accessible structure. Formatting should support reading and printing rather than decorate randomly.

Malayalam support: Word report- styles, headings, page layout, table, header/footer, page number, proofreading പാഠ്യപുസ്തകം പാഠ്യപുസ്തകം.

Study points

- Prepare a 2–3 page business report.
- Use consistent styles and check spelling, page breaks, and accessibility.

Suggested procedure

1. Create a report from a supplied outline and export a checked PDF.

Engineering / workplace application: Business reports, proposals, minutes, and procedures depend on clear document structure.

Illustrative example: Heading styles allow a report to be navigated and updated more reliably.

Common mistake: Manual formatting of every line creates inconsistency and makes revision difficult.

– M2.02 — Mail merge for personalised correspondence (3 hours)

Concept and explanation

Mail merge combines a template with a structured data source to create personalised letters or labels. Fields must map correctly, test records should be reviewed, and confidential data must be protected.

Malayalam support: Mail merge template-രീതിരേഖ data source-വിവരസ്രോതസ്സ് combine ചെയ്ത് personalised letters രൂപപ്പെടുത്തുന്നു; test records verify ചെയ്യുന്നു.

Study points

- Prepare at least five merged letters.
- Check names, addresses, dates, and blank fields.

Suggested procedure

1. Create a five-record dummy data source, merge, preview, and inspect every output.

Engineering / workplace application: Notices, invitations, invoices, and appointment letters can be generated efficiently.

Illustrative example: A name field mismatch may produce “Dear <<FirstName>>” in the final letter.

Common mistake: Sending the entire data source or an untested merged document exposes errors or privacy.

– M2.03 — Multilingual document creation using ISM (3 hours)

Concept and explanation

Multilingual documents combine correct language input, encoding, font support, layout, and proofreading. ISM and Google Typing tools support Malayalam input, but the final file must be tested on the target system. Technical terms, names, and cultural details require care.

Malayalam support: Malayalam document- input tool, encoding, font, layout, proofreading തിരുത്തൽ പരിശോധന; technical terms English- തിരുത്തൽ പരിശോധന.

Study points

- Create a bilingual document with correct typing and formatting.
- Test copy, save, print, and display on another system.

Suggested procedure

1. Type, proofread, save, reopen, and print a sample bilingual page.

Engineering / workplace application: Government, education, customer support, and local-business documents may require Malayalam and English.

Illustrative example: A bilingual notice can keep the official technical term in English and explain it in Malayalam.

Common mistake: Typing transliteration without checking the final script can change meaning.

– M2.04 — PowerPoint slide design and multimedia integration (3 hours)

Concept and explanation

A presentation uses a clear message, limited text, readable contrast, consistent layout, relevant images, and purposeful transitions. Multimedia should support the explanation and be checked for permissions, playback, and export.

Malayalam support: Slide-□ message clear, text limited, contrast readable, media relevant □□□□□□□□.

Study points

- Create a presentation with transitions.
- Export to PDF or video and verify the result.

Suggested procedure

1. Create a five-slide deck, rehearse, export, and check fonts/media.

Engineering / workplace application: Business pitches, training, project reviews, and demonstrations use presentations.

Illustrative example: A process slide can use a simple flowchart instead of a paragraph.

Common mistake: Adding effects without purpose distracts from the message.

– M2.05 — Introductory Excel data entry, formulae and charts (3 hours)

Concept and explanation

A spreadsheet should separate input data, calculations, and outputs. Formulas use cell references; absolute and relative references behave differently. Charts should match the data and question. Data validation and meaningful headings reduce errors.

Malayalam support: Spreadsheet-ഉം input, calculation, output ഉപയോഗിക്കുന്നു; cell reference, formula, chart match ഉപയോഗിക്കുന്നു.

Study points

- Create a monthly budget with correct calculations and charts.
- Check formula results with a manual sample.

Suggested procedure

1. Enter dummy monthly expenses, calculate totals/variance, and create a suitable chart.

Engineering / workplace application: Budgets, attendance, sales summaries, and inventory lists use spreadsheets.

Illustrative example: `=SUM(B2:B6)` updates when the component values change.

Common mistake: Typing calculated totals manually breaks traceability when inputs change.

– M2.06 — Series Exam I and practical record (3 hours)

Concept and explanation

Series Exam I is an official laboratory assessment item. Practical preparation includes aim, data, steps, screenshots or output, checks, and reflection for the first-half experiments.

Malayalam support: Series Exam I-രണ്ടാം ഫയൽ ഹാൻഡ്ലിംഗ്, Word, mail merge, ISM, PowerPoint, Excel revise പരീക്ഷണം.

Study points

- Organise a complete record.
- Demonstrate each tool without exposing personal data.

Suggested procedure

1. Use a timed integrated task and self-check against the rubric.

Engineering / workplace application: A structured record helps the examiner see process, output, and verification.

Illustrative example: A spreadsheet record should include input sample, formula, output, and chart interpretation.

Common mistake: Submitting only a final file without source data or checks makes errors difficult to diagnose.

Module summary: Professional office work combines formatting, multilingual access, calculation accuracy, and clear presentation.

OFFICIAL MODULE 3

Cloud Collaboration, Google Workspace and WPS Compatibility

This module develops collaborative document workflows and compatibility testing across cloud and office suites.

Hours: 12

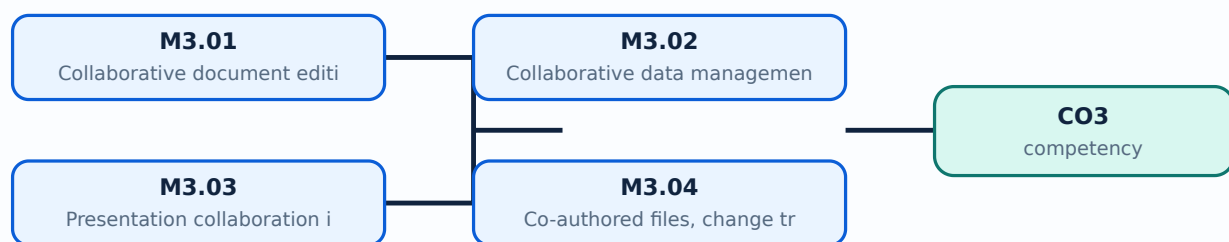
CO3 · Bloom level: Applying

Learning goals

- Edit documents collaboratively and use comments/version history.
- Manage shared data and charts in Google Sheets.
- Create collaborative presentations.
- Test WPS compatibility and complete an integrated Google Workspace project.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

– M3.01 — Collaborative document editing in Google Docs (1.5 hours)

Concept and explanation

Google Docs supports shared editing, comments, suggestions, permissions, and version history. A team should define roles, naming, review stages, and final approval.

Malayalam support: Google Docs-ൽ sharing permission, comments, suggestions, version history നോക്കാനും ഉപയോഗിക്കാനും.

Study points

- Create a 3–4 page team-edited document.
- Address comments and show version history.

Suggested procedure

1. Create a dummy document, assign roles, comment, resolve, and restore a version.

Engineering / workplace application: Policies, project reports, and meeting documents can be edited by distributed teams.

Illustrative example: Use commenter access for a reviewer and editor access for the writing team.

Common mistake: Giving edit access to everyone without role or review control can damage the document.

– M3.02 — Collaborative data management in Google Sheets (1.5 hours)

Concept and explanation

Google Sheets allows simultaneous data entry, formulas, charts, comments, and sharing. Data validation, protected ranges, version history, and clear ownership reduce errors. Charts should communicate an insight, not merely decorate.

Malayalam support: Sheets-☐ validation, protected range, version history, ownership
കാലാവസ്ഥാ പ്രവർത്തനങ്ങൾ collaborative error പ്രവർത്തനങ്ങൾ.

Study points

- Create an interactive spreadsheet with charts and team edits.
- Export to PDF/Excel and compare outputs.

Suggested procedure

1. Build a dummy sheet with roles, validation, chart, and export check.

Engineering / workplace application: Project budgets, schedules, surveys, and small dashboards use collaborative sheets.

Illustrative example: Protect a total column while allowing team members to enter input rows.

Common mistake: Concurrent editing without protected formula cells can overwrite calculations.

M3.03 — Presentation collaboration in Google Slides (1.5 hours)

Concept and explanation

Google Slides supports shared slide creation, comments, version history, media, and presentation sharing. A team should use a common theme, divide ownership, and perform a final consistency review.

Malayalam support: Slides collaboration-□ common theme, ownership, comments, history, final review □□□□□□□□.

Study points

- Prepare a professional 5–7 slide presentation.
- Include media and show team edits.

Suggested procedure

1. Assign slide sections, merge, review, rehearse, and share safely.

Engineering / workplace application: Remote project presentations and training decks benefit from simultaneous collaboration.

Illustrative example: A shared template gives every member the same title, body, and source style.

Common mistake: Different fonts, layouts, and unreviewed media make a team presentation look inconsistent.

– M3.04 — Co-authored files, change tracking and mobile usability (1.5 hours)

Concept and explanation

A co-authored set may include a report, spreadsheet, and slides connected by a common project. Change tracking and version history show responsibility. Mobile usability requires readable layout, reasonable file size, and tested links.

Malayalam support: Report, spreadsheet, slides പരസ്പരം project-പരസ്പരം link പരസ്പരം; change tracking/mobile readability test പരസ്പരം.

Study points

- Generate a co-authored set of files.
- Test on a smaller screen and verify links.

Suggested procedure

1. Create a project pack, link artefacts, inspect history, and test mobile view.

Engineering / workplace application: Business teams often move between report, data, and presentation artefacts.

Illustrative example: Use a concise table and large text for a mobile review summary.

Common mistake: A file that works on a laptop may be unreadable or inaccessible on mobile.

– M3.05 — WPS Office compatibility (1.5 hours)

Concept and explanation

Compatibility means a file opens, displays, calculates, and preserves important features across applications. Differences may occur in fonts, formulas, macros, layout, and media. Test with a representative file and document changes.

Malayalam support: WPS- file open/display/calculate/format preserve test ; fonts/formulas/media .

Study points

- Create files in different formats such as `.docx`.
- Test in another application and create a comparison chart.

Suggested procedure

1. Export a dummy document and spreadsheet, open in two suites, and record differences.

Engineering / workplace application: Interoperability supports organisations using different office suites.

Illustrative example: A formula may display differently if a function or locale is unsupported.

Common mistake: Assuming an extension guarantees identical behaviour ignores application differences.

– M3.06 — End-to-end collaboration in Google Workspace (3 hours)

Concept and explanation

An integrated workspace project connects Docs, Sheets, Slides, Drive sharing, comments, and version history around a business task such as event planning. The full cycle includes plan, data collection, analysis, report, presentation, review, approval, and archive.

Malayalam support: Integrated project- Drive, Docs, Sheets, Slides, sharing, review, approval, archive cycle .

Study points

- Complete an event-planning workflow.
- Document access, changes, decisions, and final outputs.

Suggested procedure

1. Plan → collect → calculate → document → present → review → approve → archive.

Engineering / workplace application: Event planning, project tracking, and small-business operations use connected cloud tools.

Illustrative example: A budget sheet can feed a report and a presentation after controlled review.

Common mistake: Creating separate files without a shared naming/access plan causes duplication.

Module summary: Collaboration is a process of roles, permissions, version history, review, compatibility, and shared accountability.

OFFICIAL MODULE 4

AI Content, Editing, Spreadsheet Automation and Open-ended Work

This module uses AI tools for office tasks while preserving verification, ethics, privacy, and a reflective practical record.

Hours: 12

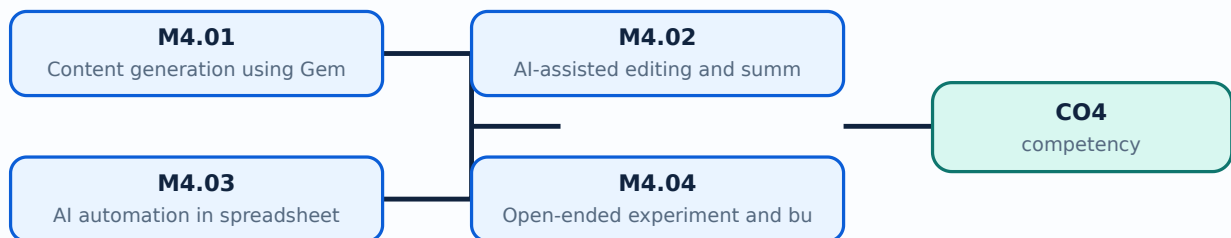
CO4 · Bloom level: Applying

Learning goals

- Generate and edit content with AI tools.
- Use AI-assisted editing and summarisation.
- Automate spreadsheet and presentation tasks.
- Complete an open-ended experiment with prompt and result records.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

– M4.01 — Content generation using Gemini or ChatGPT (3 hours)

Concept and explanation

Generative AI can draft text, outlines, tables, ideas, and summaries from prompts. Users should provide context and constraints, avoid confidential data, verify facts, edit for audience and originality, and record prompts and outputs.

Malayalam support: AI draft useful കൂടുതൽ context/constraints കൂടുതൽ; confidential data ഒഴിവാക്കുക output verify/edit ചെയ്യുക.

Study points

- Create AI-generated content, edit it, and insert it into a document.
- Include ethical considerations and prompt record.

Suggested procedure

1. Use dummy data, save prompt/output, verify claims, edit, and cite or acknowledge use as required.

Engineering / workplace application: AI can accelerate routine drafting while the student remains responsible for accuracy and appropriate use.

Illustrative example: Ask for a neutral business memo with a word limit, then fact-check and revise it.

Common mistake: Copying AI output without checking facts, sources, bias, or originality is unacceptable.

M4.02 — AI-assisted editing and summarisation with Grammarly or Copilot (3 hours)

Concept and explanation

AI editing tools can suggest grammar, clarity, tone, and summaries. Suggestions must be accepted selectively because a tool may change meaning, names, technical terms, or culturally important language. A change record demonstrates judgement.

Malayalam support: AI suggestion  accept ; meaning, technical term, name, Malayalam context  verify .

Study points

- Create a formatted document using suggestions.
- Record changes made and explain rejected suggestions.

Suggested procedure

1. Run a dummy document through editing support, compare versions, and record decisions.

Engineering / workplace application: Editing support improves a draft when the author reviews every important change.

Illustrative example: A technical sentence may need an English term even when a simpler synonym is suggested.

Common mistake: Optimising grammar can remove the writer's intended meaning or evidence.

Concept and explanation

AI may generate formulas, classify rows, suggest charts, summarise trends, or propose slide content. Automation should be tested with known cases, and the prompt, data, formula, output, and human verification should be recorded.

Malayalam support: AI formula/chart output known cases തിരിച്ചറിയുക test ചെയ്യുക; prompt, data, result, verification record രേഖപ്പെടുത്തുക.

Study points

- Generate an automated spreadsheet or presentation section.
- Check outputs manually and record limitations.

Suggested procedure

1. Create a dummy dataset, prompt the tool, test three cases, and document result.

Engineering / workplace application: AI-assisted spreadsheet analysis can help a small office identify trends or anomalies.

Illustrative example: Verify an AI-generated total against a hand calculation before using the chart.

Common mistake: An incorrect formula repeated across rows creates systematic error.

– M4.04 — Open-ended experiment and business sustainability (3 hours)

Concept and explanation

The open-ended experiment allows a student or group to extend an existing task, integrate tools, design a practical workflow, or study comparative performance. The source asks for a portfolio/reflection on tool efficiency, challenges, business sustainability, and individual/team management, although one sentence is corrupted in the PDF.

Malayalam support: Open-ended project- tool efficiency, challenge, sustainability, teamwork, reflection .

Study points

- Choose a practical office-automation problem.
- Submit aim, workflow, outputs, prompts, verification, reflection, and limitations.

Suggested procedure

1. Define baseline → design → test → compare → reflect → present; use dummy data.

Engineering / workplace application: Reducing duplicate work, paper use, or unnecessary re-entry can support sustainable office practice.

Illustrative example: Compare manual meeting-summary preparation with an AI-assisted draft plus human verification.

Common mistake: Claiming that AI saved time without measuring baseline and actual time is weak evidence.

– M4.05 — Series Exam II and lab examination preparation (3 hours)

Concept and explanation

Series Exam II and the final practical examination require integrated performance across file management, office tools, collaboration, and responsible AI. The record should be complete, organised, and reproducible.

Malayalam support: Series Exam II/lab exam-ഫയലുകൾ, ഓഫീസ്, ക്ലൗഡ്, AI വർക്ക്ഫ്ലോസ് ഇന്റഗ്രേറ്റഡ് രീതിയിൽ എഡിറ്റ് ചെയ്യുക.

Study points

- Practise a timed integrated task.
- Check permissions, output, record, and viva explanation.

Suggested procedure

1. Run a mock practical from folder creation through final presentation and archive.

Engineering / workplace application: A final practical can evaluate not only the final file but also safe and reproducible method.

Illustrative example: A student should be able to explain where a file came from, how it was changed, and how output was checked.

Common mistake: Leaving screenshots, prompts, or verification until the last day produces incomplete records.

Module summary: AI-assisted productivity is valuable only when the student verifies output, protects data, records decisions, and reflects on impact.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

The diagram library consists of five horizontal bars, each representing a module. The first bar is labeled 'Module 1: Introduction to AI'. The second bar is labeled 'Module 2: MS Office'. The third bar is labeled 'Module 3: Cloud'. The fourth bar is labeled 'Module 4: AI Content'. The fifth bar is labeled 'Module 5: AI Ethics'. Each bar has a corresponding button on the right side with the module name and a link icon. The buttons are labeled 'Module 2: MS Office', 'Module 3: Cloud', and 'Module 4: AI Content'. Below the bars, there is a link that says 'Open the module to view its labelled learning-map diagram.'

[Module 2: MS Office](#)

[Module 3: Cloud](#)

[Module 4: AI Content,](#)

[Open the module to view its labelled learning-map diagram.](#)

Official Activities and Practical Requirements

Official activities / self-learning

- Weeks 1-15, 1.5 hours per week, total self-learning 22.5 hours.
- Prepare rough and fair records for the listed experiments.
- Prepare for laboratory examination and open-ended activity.
- Use screenshots, version history, outputs, and reflective notes in the practical record.

Practical requirements / equipment

- 30 Intel i5 or higher computers/laptops with minimum 8 GB RAM, 256 GB SSD, numeric keypad, webcam, and microphone.
- 50 Mbps or above high-speed internet; one short-throw multimedia projector; one shared laser/inkjet printer.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Attendance 5; continuous evaluation converted to 30; practical tests converted to 15 + 15; open-ended experiment converted to 10; practical ESE 40; practical total 100.
- CIE rubric total 50: preparation/punctuality 10; setup/procedure 10; observation/data 5; analysis/result 10; viva/concept 10; workmanship/discipline/teamwork 5. Practical test/ESE rubric total 40: write-up/procedure 10; setup/execution 10; observation/result/output 8; viva 8; record 4. Open-ended rubric total 50.
- The open-ended section contains an obvious OCR/layout corruption combining “Modify or extend an experiment” with portfolio/reflection text. Repeated or misaligned experiment numbers in the self-learning table are preserved as source anomalies rather than silently corrected.

Module-wise Questions and Answers

module-1 — File Creation, Organisation, Security and Recovery

1. Explain File creation and exploration. [3 marks]

Files store documents, data, media, and program outputs. Students should identify file type, extension, location, owner, version, and purpose. Naming conventions use meaningful names, dates, and version labels without unsafe characters or ambiguity.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Basic file organisation. [3 marks]

Folder structures should reflect business function, project, date, and access need. A documented navigation path helps another user find a record without guesswork. Avoid excessive nesting and duplicate copies.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain File compression and archiving. [3 marks]

Compression reduces storage or transfer size and archiving preserves a project snapshot. Integrity checks confirm that the archive opens and files are complete. The original should be retained according to policy.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain File permissions and access control. [3 marks]

Access control determines who may read, modify, share, or delete a file. Least privilege, strong authentication, secure sharing, and access logs protect privacy. Permissions should match role and be reviewed.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — MS Office Documents, Presentations, Malayalam Typing and Excel

5. Explain MS Word document creation and formatting. [3 marks]

A professional Word report uses styles, headings, page layout, tables, headers/footers, page numbers, consistent spacing, proofreading, and accessible structure. Formatting should support reading and printing rather than decorate randomly.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Mail merge for personalised correspondence. [3 marks]

Mail merge combines a template with a structured data source to create personalised letters or labels. Fields must map correctly, test records should be reviewed, and confidential data must be protected.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Multilingual document creation using ISM. [3 marks]

Multilingual documents combine correct language input, encoding, font support, layout, and proofreading. ISM and Google Typing tools support Malayalam input, but the final file must be tested on the target system. Technical terms, names, and cultural details require care.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain PowerPoint slide design and multimedia integration. [3 marks]

A presentation uses a clear message, limited text, readable contrast, consistent layout, relevant images, and purposeful transitions. Multimedia should support the explanation and be checked for permissions, playback, and export.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Cloud Collaboration, Google Workspace and WPS Compatibility

9. Explain Collaborative document editing in Google Docs. [3 marks]

Google Docs supports shared editing, comments, suggestions, permissions, and version history. A team should define roles, naming, review stages, and final approval.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Collaborative data management in Google Sheets. [3 marks]

Google Sheets allows simultaneous data entry, formulas, charts, comments, and sharing. Data validation, protected ranges, version history, and clear ownership reduce errors. Charts should communicate an insight, not merely decorate.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Presentation collaboration in Google Slides. [3 marks]

Google Slides supports shared slide creation, comments, version history, media, and presentation sharing. A team should use a common theme, divide ownership, and perform a final consistency review.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Co-authored files, change tracking and mobile usability. [3 marks]

A co-authored set may include a report, spreadsheet, and slides connected by a common project. Change tracking and version history show responsibility. Mobile usability requires readable layout, reasonable file size, and tested links.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — AI Content, Editing, Spreadsheet Automation and Open-ended Work

13. Explain Content generation using Gemini or ChatGPT. [3 marks]

Generative AI can draft text, outlines, tables, ideas, and summaries from prompts. Users should provide context and constraints, avoid confidential data, verify facts, edit for audience and originality, and record prompts and outputs.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain AI-assisted editing and summarisation with Grammarly or Copilot. [3 marks]

AI editing tools can suggest grammar, clarity, tone, and summaries. Suggestions must be accepted selectively because a tool may change meaning, names, technical terms, or culturally important language. A change record demonstrates judgement.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain AI automation in spreadsheets and presentations. [3 marks]

AI may generate formulas, classify rows, suggest charts, summarise trends, or propose slide content. Automation should be tested with known cases, and the prompt, data, formula, output, and human verification should be recorded.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Open-ended experiment and business sustainability. [3 marks]

The open-ended experiment allows a student or group to extend an existing task, integrate tools, design a practical workflow, or study comparative performance. The source asks for a portfolio/reflection on tool efficiency, challenges, business sustainability, and individual/team management, although one sentence is corrupted in the PDF.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. Define or explain *File creation and exploration* with one relevant application. (5 marks)
2. Define or explain *Basic file organisation* with one relevant application. (5 marks)
3. Define or explain *MS Word document creation and formatting* with one relevant application. (5 marks)
4. Define or explain *Mail merge for personalised correspondence* with one relevant application. (5 marks)
5. Define or explain *Collaborative document editing in Google Docs* with one relevant application. (5 marks)

6. **6.** Define or explain *Collaborative data management in Google Sheets* with one relevant application. (5 marks)
7. **7.** Define or explain *Content generation using Gemini or ChatGPT* with one relevant application. (5 marks)
8. **8.** Define or explain *AI-assisted editing and summarisation with Grammarly or Copilot* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **File Creation, Organisation, Security and Recovery:** Good office automation begins with disciplined file names, structure, access, backup, and retrieval.
- **MS Office Documents, Presentations, Malayalam Typing and Excel:** Professional office work combines formatting, multilingual access, calculation accuracy, and clear presentation.
- **Cloud Collaboration, Google Workspace and WPS Compatibility:** Collaboration is a process of roles, permissions, version history, review, compatibility, and shared accountability.
- **AI Content, Editing, Spreadsheet Automation and Open-ended Work:** AI-assisted productivity is valuable only when the student verifies output, protects data, records decisions, and reflects on impact.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
File creation and exploration	Files store documents, data, media, and program outputs. Students should identify file type, extension, location, owner, version, and purpose. Naming conventions use meaningful names, dates, and version labels without unsafe characters or ambiguity.
Basic file organisation	Folder structures should reflect business function, project, date, and access need. A documented navigation path helps another user find a record without guesswork. Avoid excessive nesting and duplicate copies.
File compression and archiving	Compression reduces storage or transfer size and archiving preserves a project snapshot. Integrity checks confirm that the archive opens and files are complete. The original should be retained according to policy.
File permissions and access control	Access control determines who may read, modify, share, or delete a file. Least privilege, strong authentication, secure sharing, and access logs protect privacy. Permissions should match role and be reviewed.
Backup and recovery strategies	Backup copies support recovery from deletion, corruption, hardware failure, or ransomware. A plan states what is backed up, frequency, location, retention, access, and test-recovery procedure. Recovery should be tested rather than assumed.
File search and filtering techniques	Search uses file name, type, date, size, content, tags, or location to find records. Filters reduce a large set to relevant files. Search results should be checked for version, authority, and access.
MS Word document creation and formatting	A professional Word report uses styles, headings, page layout, tables, headers/footers, page numbers, consistent spacing, proofreading, and accessible structure. Formatting should support reading and printing rather than decorate randomly.
Mail merge for personalised correspondence	Mail merge combines a template with a structured data source to create personalised letters or labels. Fields must map correctly, test records should be reviewed, and confidential data must be protected.
Multilingual document creation using ISM	Multilingual documents combine correct language input, encoding, font support, layout, and proofreading. ISM and Google Typing tools support Malayalam input, but the final file must be tested on the target system. Technical terms, names, and cultural details r
PowerPoint slide design and multimedia integration	A presentation uses a clear message, limited text, readable contrast, consistent layout, relevant images, and purposeful transitions. Multimedia should support

Term	Short meaning
	the explanation and be checked for permissions, playback, and export.
Introductory Excel data entry, formulae and charts	A spreadsheet should separate input data, calculations, and outputs. Formulas use cell references; absolute and relative references behave differently. Charts should match the data and question. Data validation and meaningful headings reduce errors.
Series Exam I and practical record	Series Exam I is an official laboratory assessment item. Practical preparation includes aim, data, steps, screenshots or output, checks, and reflection for the first-half experiments.
Collaborative document editing in Google Docs	Google Docs supports shared editing, comments, suggestions, permissions, and version history. A team should define roles, naming, review stages, and final approval.
Collaborative data management in Google Sheets	Google Sheets allows simultaneous data entry, formulas, charts, comments, and sharing. Data validation, protected ranges, version history, and clear ownership reduce errors. Charts should communicate an insight, not merely decorate.
Presentation collaboration in Google Slides	Google Slides supports shared slide creation, comments, version history, media, and presentation sharing. A team should use a common theme, divide ownership, and perform a final consistency review.
Co-authored files, change tracking and mobile usability	A co-authored set may include a report, spreadsheet, and slides connected by a common project. Change tracking and version history show responsibility. Mobile usability requires readable layout, reasonable file size, and tested links.
WPS Office compatibility	Compatibility means a file opens, displays, calculates, and preserves important features across applications. Differences may occur in fonts, formulas, macros, layout, and media. Test with a representative file and document changes.
End-to-end collaboration in Google Workspace	An integrated workspace project connects Docs, Sheets, Slides, Drive sharing, comments, and version history around a business task such as event planning. The full cycle includes plan, data collection, analysis, report, presentation, review, approval, and arch
Content generation using Gemini or ChatGPT	Generative AI can draft text, outlines, tables, ideas, and summaries from prompts. Users should provide context and constraints, avoid confidential data, verify facts, edit for audience and originality, and record prompts and outputs.
AI-assisted editing and summarisation with Grammarly or Copilot	AI editing tools can suggest grammar, clarity, tone, and summaries. Suggestions must be accepted selectively because a tool may change meaning, names, technical terms, or culturally important language. A change record demonstrates judgement.
AI automation in spreadsheets and presentations	AI may generate formulas, classify rows, suggest charts, summarise trends, or propose slide content. Automation should be tested with known cases, and the prompt, data, formula, output, and human verification should be recorded.
Open-ended experiment and business sustainability	The open-ended experiment allows a student or group to extend an existing task, integrate tools, design a practical workflow, or study comparative performance. The source asks for a portfolio/reflection on tool efficiency, challenges, business sustainability,

Term	Short meaning
Series Exam II and lab examination preparation	Series Exam II and the final practical examination require integrated performance across file management, office tools, collaboration, and responsible AI. The record should be complete, organised, and reproducible.

Official References and Resources

References listed in the supplied syllabus

1. Microsoft Office 2019 Step by Step, Joan Lambert and Curtis Frye, Microsoft Press.
2. Understanding Google Docs: The Step-by-step Guide to Understanding the Fundamentals of Google Docs, Kevin Wilson, Elluminet Press.
3. ISM Malayalam User Manual, CDAC.
4. The New Microsoft Office 365 Bible, Rick Sterling and Mark Finch, Techmind Press LLC.

Official learning resources listed

- <https://www.usaii.org/ai-insights/8-best-ai-tools-for-small-business-leaders>
- <https://bernardmarr.com/20-game-changing-ai-tools-every-small-business-leader-needs-now/>

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